Effects of Duration of Early Deprivation on Brain Development in Post-Institutionalized Adolescents

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Introduction

Animal studies have documented neuroanatomical and behavioral effects of early life deprivation. Studies of children adopted from institutional care (**post-institutionalized; PI children**) suggest that early deprivation is associated with:

- differences in *cognitive and socioemotional development* (Gunnar et al., 2000)
- changes in the volume of the brain's *limbic system* (Mehta et al., 2009; Tottenham et al., 2010)
- altered white matter development of the brain's *prefrontal cortex* (Eluvathingal et al., 2006; Behen et al., 2009; Govindan et al., 2010)

Although individual variability exists, **longer duration of institutional care** is predictive of reduced physical catch-up and poorer cognitive achievement. (Van IJzendoorn et al. 2007; Johnson, 2002)



http://www.evolutionaryparenting.com/wp-content/uploads/2011/08/cribs_-_babies_in_orphanage.jpg

Few studies have investigated the relationship between duration of institutional care and structural brain development in PI adolescents. Based on animal models of early life stress (e.g. Vyas et al., 2002), we hypothesized that longer exposure to institutional care would be associated with increased amygdala volume and decreased hippocampal and prefrontal volumes as measured at adolescence.

Questions

Do post-institutionalized (PI) children show altered development of limbic and prefrontal structures?

Are changes in brain volume more pronounced in later-adopted PI children?

Participants

12-14 year old children either adopted internationally from institutional care or raised in Minnesota with their biological family

Controls	Earlier Adopted (PI-EA)	Later Adopted (PI-LA)
N = 62 *(32 females)*	N = 56 *(43 females)*	N = 54 *(30 females)*
Not adopted	Adopted before 12 months	Adopted between 13-78 months
• No developmental, neurological, or psychiatric disorders	• No FAS or developmental disorders • At least 50% of pre-adoptive care spent in an institution • Diverse countries of origin	

Demographics: Both PI and control children lived primarily in two-parent families, with most households having at least one parent who had completed college or a graduate level degree. Unlike PI children, control children were predominantly Caucasian (88%).



Excluded Participants: The final data set described above reflects a subset of participants from a larger project who provided segmented structural imaging data of acceptable quality.

Methods

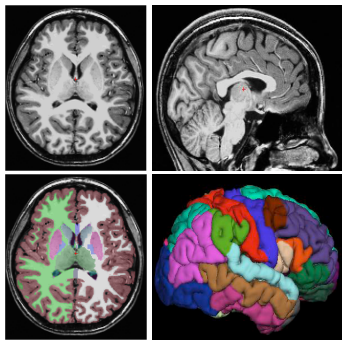
***Structural MRI Scan*:** T1-weighted 3D MPRAGE anatomical series acquired on a Siemens 3T Trio Scanner

- TR = 2530 ms, TE = 3.56 ms, FOV = 256 mm, flip angle = 7 degrees
- 1 mm iso-voxel, 240 sagittal slices

MRI Analyses: Freesurfer Image Analysis Suite was used to obtain automated, volumetric segmentation data

Initial analyses included age, sex, and total intracranial volume (ICV; a proxy for head size) as covariates

Follow-up analyses included age, sex, and adjusted supratentorial volume (adjSTV; a more stringent correction factor specific to cortical and subcortical brain tissue) as covariates



Cortical Gray Matter Volume

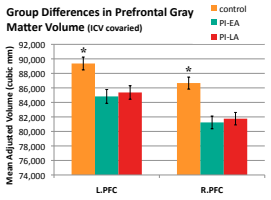
	F	*p*	$\eta^2_{partial}$	PI-EA *B* (*se*)	PI-LA *B* (*se*)
Frontal Lobe					
left	8.286	.000*	.090	-5740.10 (1609.91)*	-5233.61 (1528.93)*
right	11.749	.000*	.124	-6604.17 (1534.92)*	-5893.71 (1457.71)*
Occipital Lobe					
left	5.359	.006*	.061	-1682.01 (514.04)*	-821.03 (488.18)
right	3.709	.027*	.043	-1393.22 (515.03)*	-507.20 (489.12)
Parietal Lobe					
left	5.225	.006*	.059	-3582.13 (1188.60)*	-2807.62 (1128.81)*
right	5.241	.006*	.059	-3278.69 (1196.27)*	-3224.75 (1136.09)*
Temporal Lobe					
left	6.368	.006*	.061	-2131.20 (962.97)	-2914.76 (914.52)*
right	9.023	.000*	.098	-3149.62 (994.44)*	-3790.48 (944.42)*

Comparison of age and sex adjusted cortical lobular volumes (mm^3) (corrected for total intracranial volume).

B and associated standard errors give the parameter estimates for the contrast of PI-EA and PI-LA in comparison to controls.

- **PI children had smaller volumes across all lobes of the brain** vs. non-adopted children, suggesting gross differences in brain tissue growth relative to head size
- Follow-up analyses remained significant only in right prefrontal cortex, suggesting **prefrontal cortex may be especially sensitive to effects of early adversity**

Prefrontal Cortex Volume



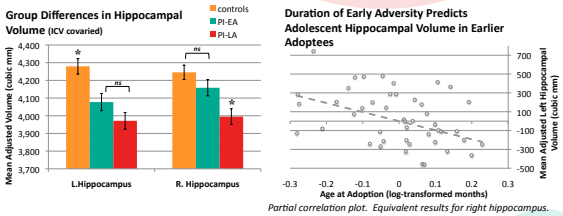
- **Both EA and LA children had smaller prefrontal gray matter volumes**
- There was no linear relationship between age at adoption and prefrontal volume
- Volumetric decreases in prefrontal cortex for PI children were driven mostly by **reductions in prefrontal surface area**

Acknowledgments

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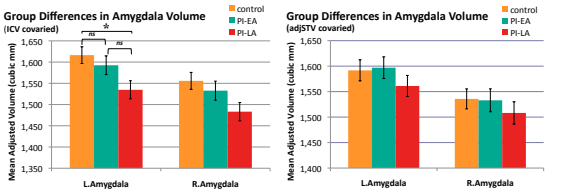
Hippocampal Volume



Partial correlation plot. Equivalent results for right hippocampus.

- **Both EA and LA children had smaller hippocampal volumes, with greater volume reductions in the LA group** *(follow-up analyses were also significant)*
- Similarly, older age at adoption predicted smaller hippocampal volumes even within children adopted prior to 12 months of age

Amygdala Volume



- **LA children had smaller left amygdala volumes than non-adopted children, although this effect did not survive more stringent follow-up analyses**

Discussion

We found evidence for long-term effects of early institutional care on global brain growth, even after controlling for variability in head size

PI children showed altered structural brain development in brain regions implicated in animals models of early life stress:

- *Prefrontal cortex volume* was reduced in both EA and LA children, suggesting that this region is especially vulnerable to early adversity
- *Hippocampal volumes* showed evidence of dose-dependent effects of early life stress
- Unlike previous studies, no *amygdala volume* differences were detected

We did not find strong evidence for the duration of early life stress impacting brain development, given the largest differences in structural brain development were between PI children and non-adopted controls

Future studies should investigate the functional implications of volumetric reductions in key brain regions in PI children, as well as their relation to growth-stunting, catch-up growth, and altered pubertal timing in this population

Conclusion

Early adversity has lasting impacts on the structural development of specific regions of the human brain into the adolescent years.